

International University

School of Computer Science & Engineering

Web Application Development

IT093IU

Final Report

Dining Guide

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Date Performed: 19/02/2025

Date Submitted: 31/05/2025

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CHAPTER I

INTRODUCTION

I. Overview

1. About us

Dining Guide is a community-driven platform where food lovers can share honest reviews and discover great restaurants. What started as a personal blog to document one foodie's dining adventures has grown into a space where anyone can contribute their own experiences. Whether you're hunting for hidden gems or sharing your latest food find, our mission is to connect people through good food and authentic recommendations.

Dining Guide Github can be accessed by clicking [here](#).

2. Motivation

The motivation behind this project was to create a simple yet meaningful platform for sharing dining experiences. As a team who enjoy exploring new restaurants, we wanted a way to document my visits and help others discover great places to eat. The goal is to build a trustworthy and engaging space where users can find genuine reviews, share their own food stories, and connect over a shared love of dining out.

3. Scope

This project focuses on developing a web-based dining guide that allows users to browse, write, and read restaurant reviews. The platform includes core features such as restaurant listings, user-generated reviews with ratings and images, and a simple authentication system. It supports both personal and community contributions, offering a user-friendly interface for discovering new dining spots. The current scope covers essential functionalities for review posting and display, with plans for future enhancements like AI support chat.

4. Target users

The Dining Guide is designed for a wide range of users who enjoy exploring food and dining experiences. This includes:

- **Food enthusiasts** who want to share their personal reviews and discover new restaurants.
- **Travelers and locals** looking for trusted, user-generated recommendations in different areas.
- **Casual diners** who want to read real opinions before choosing a place to eat.

The platform is intended to be accessible and easy to use for anyone interested in food, regardless of their technical background.

II. Member contribution and Work breakdown

1. Member and contribution

Table 1 - Member and occupation

Name	Student ID	Occupation
Phan An Phú	ITITIU21280	Leader
Lê Quan Phát Thành	ITITIU21312	Member
Lê Phương Nghi	ITITIU21253	Member

Table 2 - Individual responsibility and contribution

Full Name	Responsibility	Contribution
Phan An Phú	• Designed and implemented the server-side logic using Node.js and Express. • Set up and managed the database (MySQL), including schema design for users, reviews, and restaurants. • Handled API development for CRUD operations (create, read, update, delete reviews/restaurants).	36%

	• Implemented user authentication and session management. • Managed project structure, Git repository, and coordinated tasks across the team. • Led team meetings and decision-making on key features and architecture.	
Lê Quan Phát Thành	• Developed the user interface using React. • Designed and implemented pages for home, restaurant detail, login/signup, and review submission. • Focused on responsive design for both mobile and desktop views. • Worked closely with the backend to connect API data with UI components. • Created reusable components for things like review cards, navigation bars, and modals. • Handled form validation and user interaction features.	33%
Lê Phương Nghi	• Helped define the project scope, feature list, and timeline. • Researched similar platforms and gathered ideas for improvement. • Created wireframes and simple mockups during the planning phase. • Wrote documentation, report drafts, and presentation slides. • Supported testing, reviewed user experience, and gave feedback to improve usability. • Assisted in preparing demo content and sample data for testing.	31%

2. Work breakdown structure

The structure of the project can be expressed below:

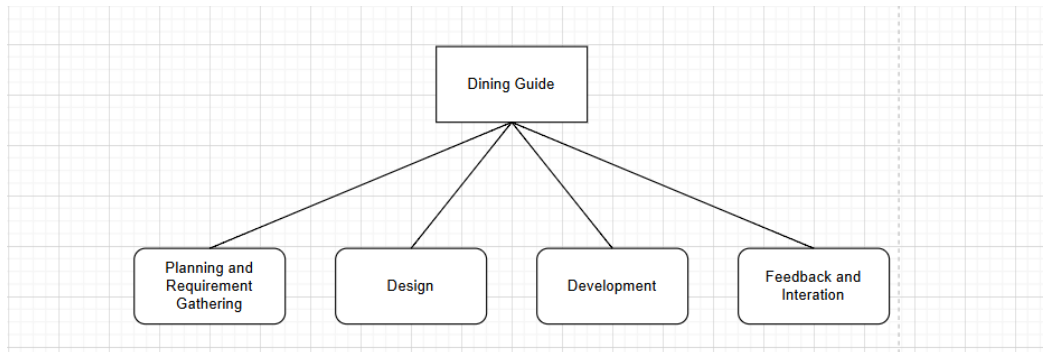


Figure 1 – Project Overall Structure

The tasks for the Planning and Requirement Gathering is described below:

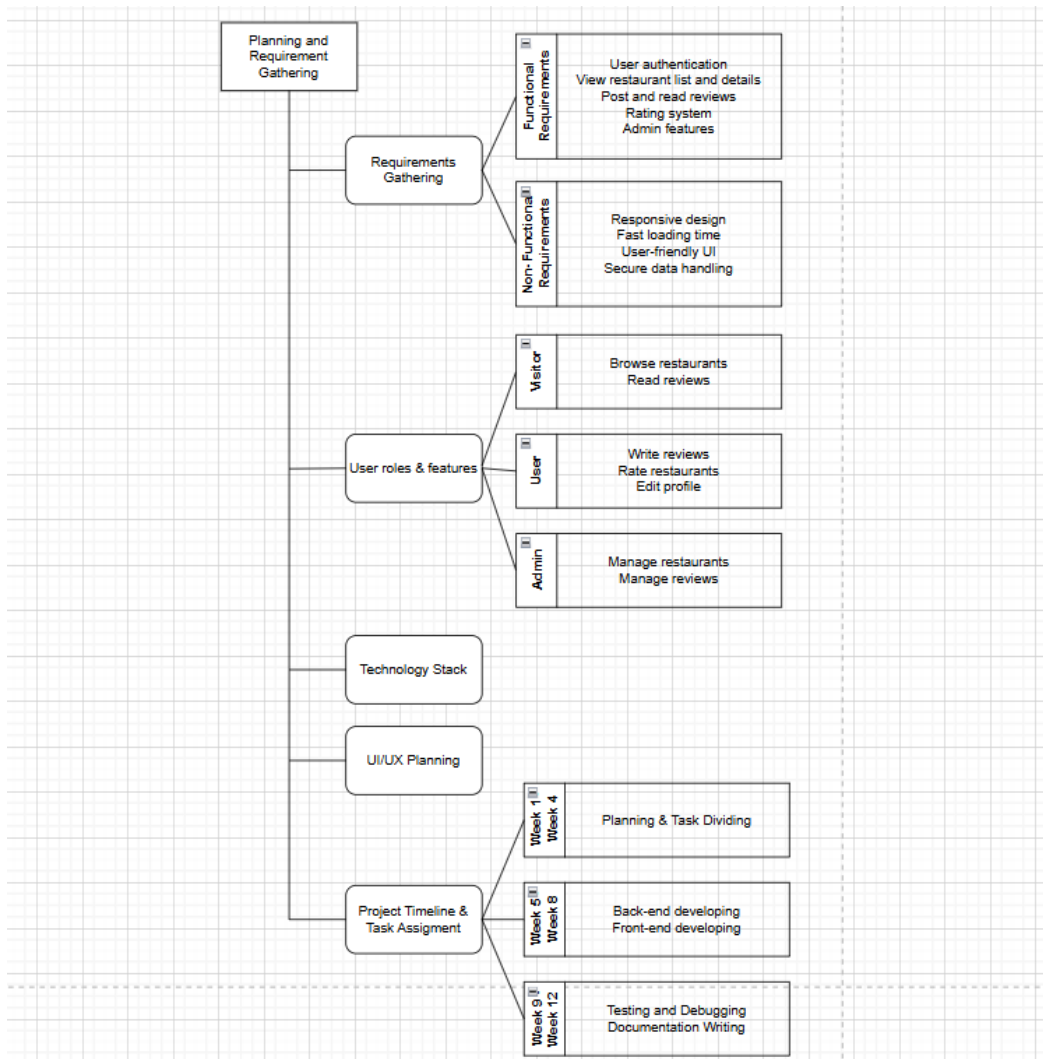


Figure 2 – Planning and Requirement Gathering stage tree

The tasks for the Design is described below:

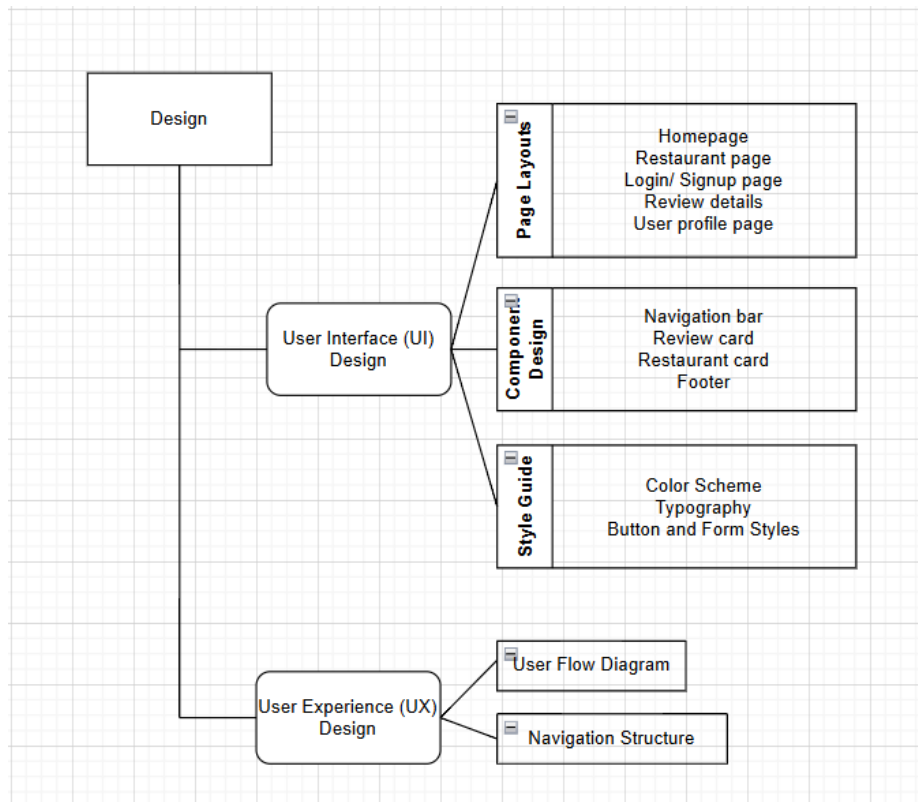


Figure 3 – Design stage tree

The tasks for the Development is described below:

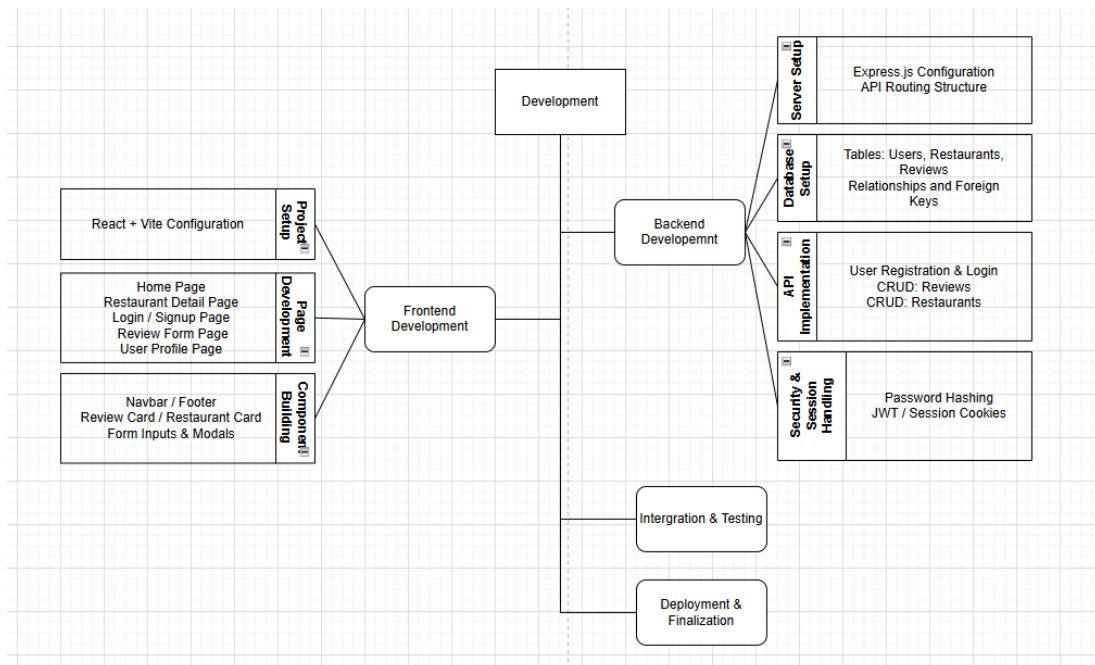


Figure 4 – Development stage tree

The tasks for the Feedback & Iteration is described below:

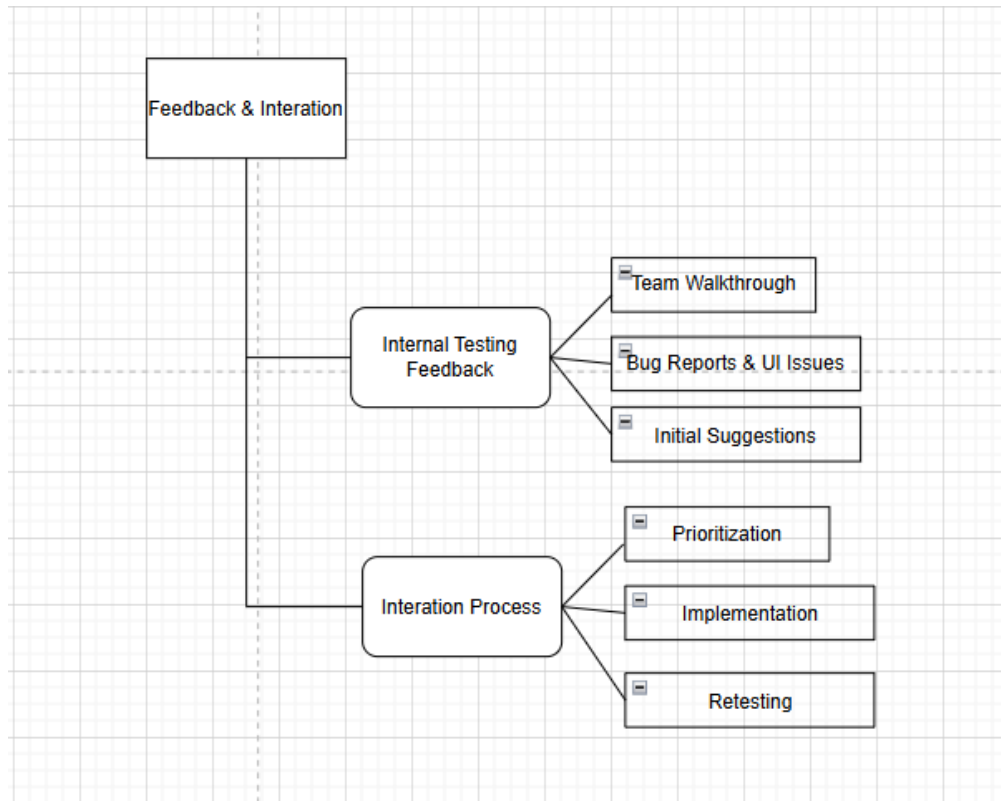


Figure 5 – Feedback & Iteration stage tree

CHAPTER II

PROJECT ANALYSIS

I. Requirement Analysis

1. Interface Requirement

The Dining Guide web application should offer a clean, user-friendly interface that allows both guests and registered users to easily navigate and interact with restaurant content. By using available designing website (Figma), we designed an expecting pages. Key interface requirements include:

- **Responsive Design:** The interface must adapt seamlessly to various screen sizes, including desktop, tablet, and mobile devices.
- **Clear Navigation:** A consistent header and footer should provide access to the home page, restaurant listings, login/signup, and user profile.
- **Interactive Components:** Users should be able to submit reviews, rate restaurants, and browse others' feedback with clearly styled forms and buttons.
- **Visual Consistency:** The UI must maintain a cohesive look with a defined color scheme, readable typography, and intuitive iconography.
- **Feedback Mechanisms:** Users should receive real-time feedback for actions (e.g., form validation messages, loading spinners, and success alerts).
- **Accessibility:** Basic accessibility features such as alt text for images, keyboard navigability, and adequate contrast ratios should be followed.

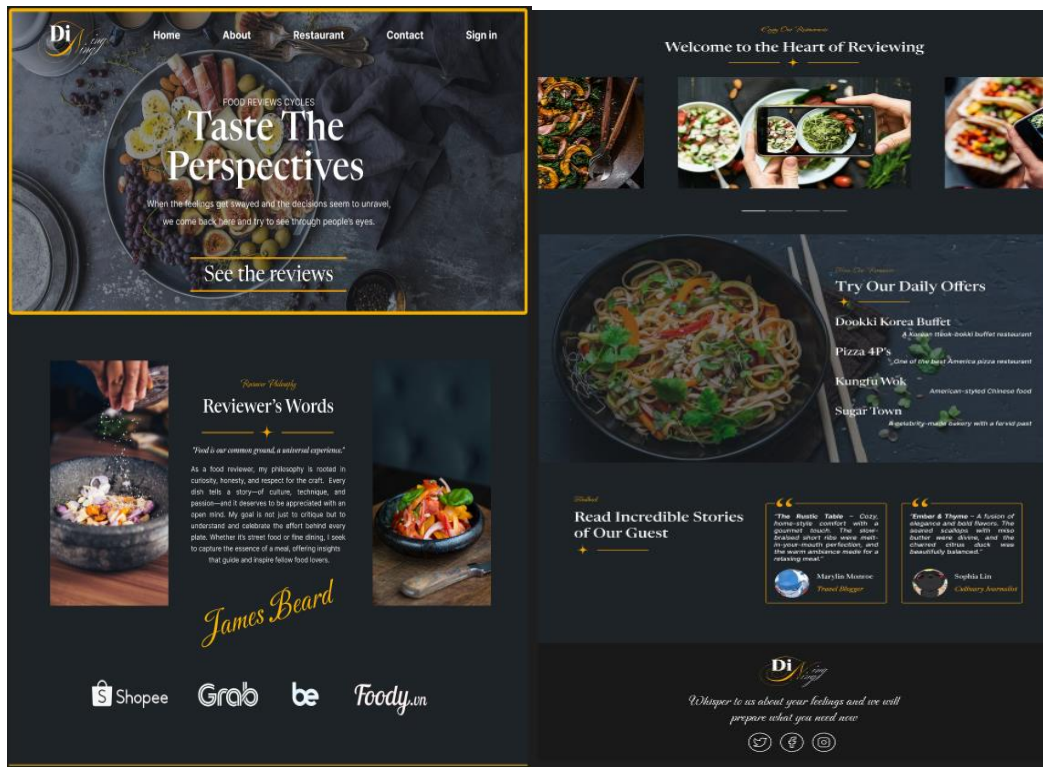


Figure 6 – Expected Homepage

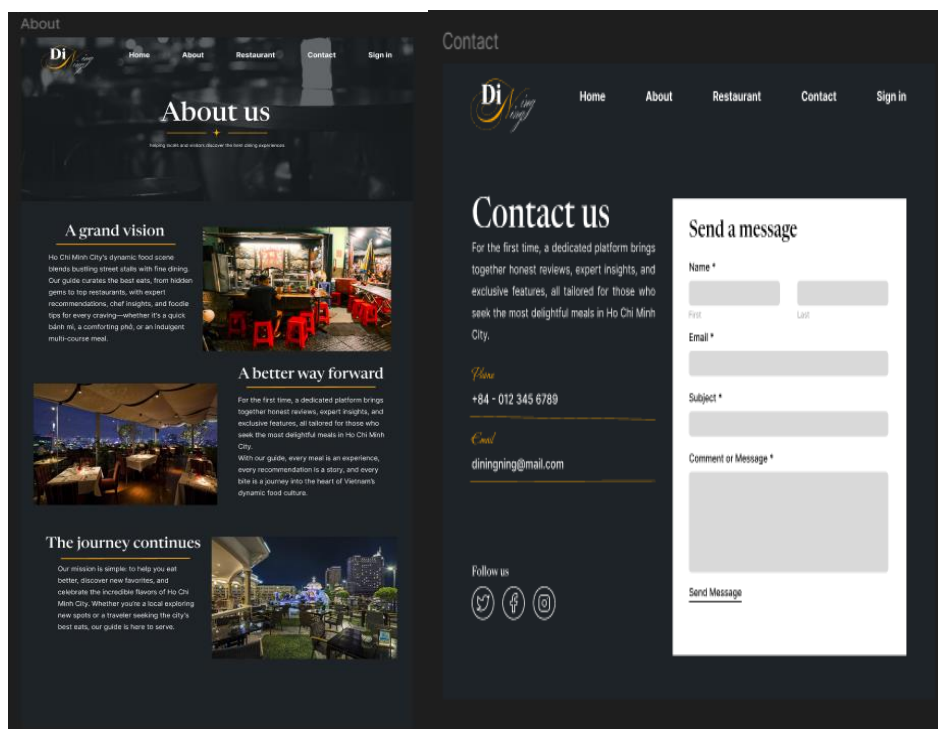


Figure 7 – Expected About us and Contact us pages

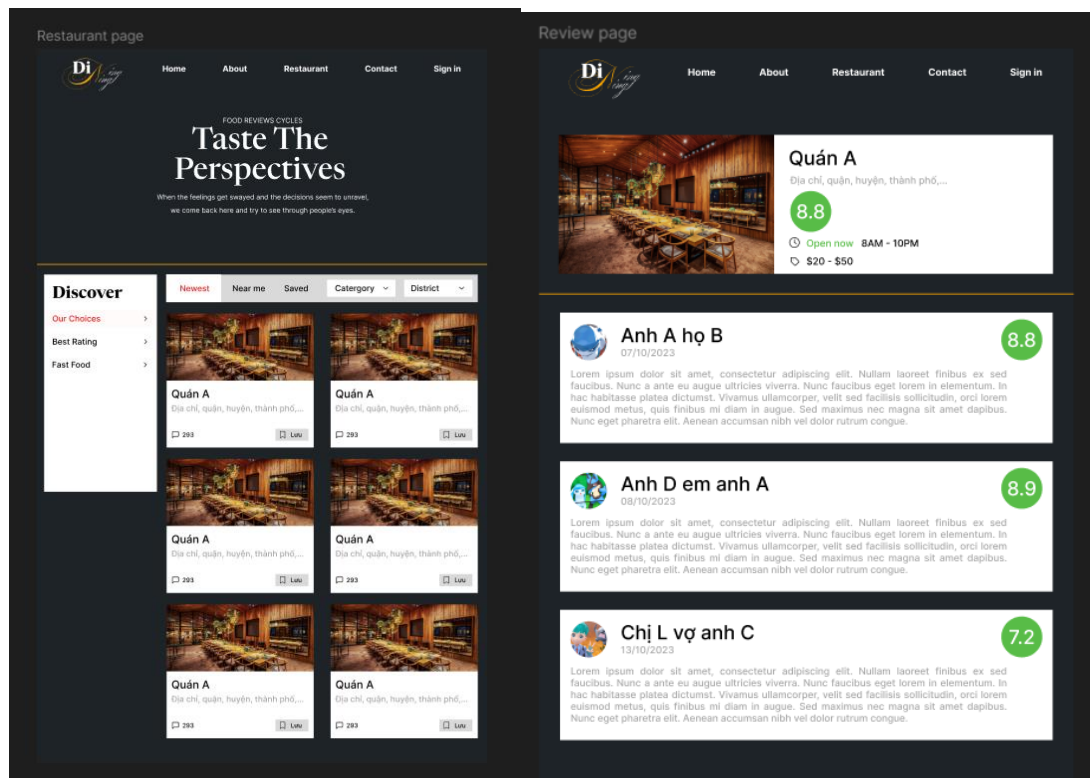


Figure 8 – Expected Restaurant and Review pages

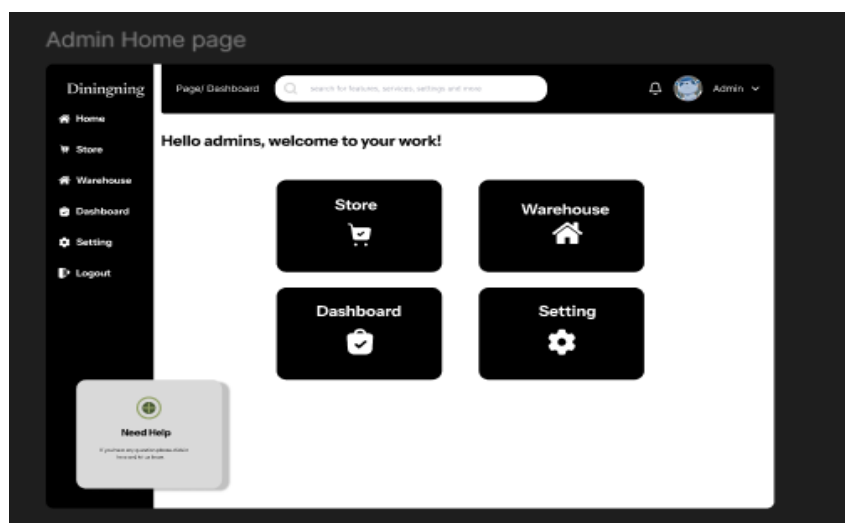


Figure 9 – Expected Admin Homepage

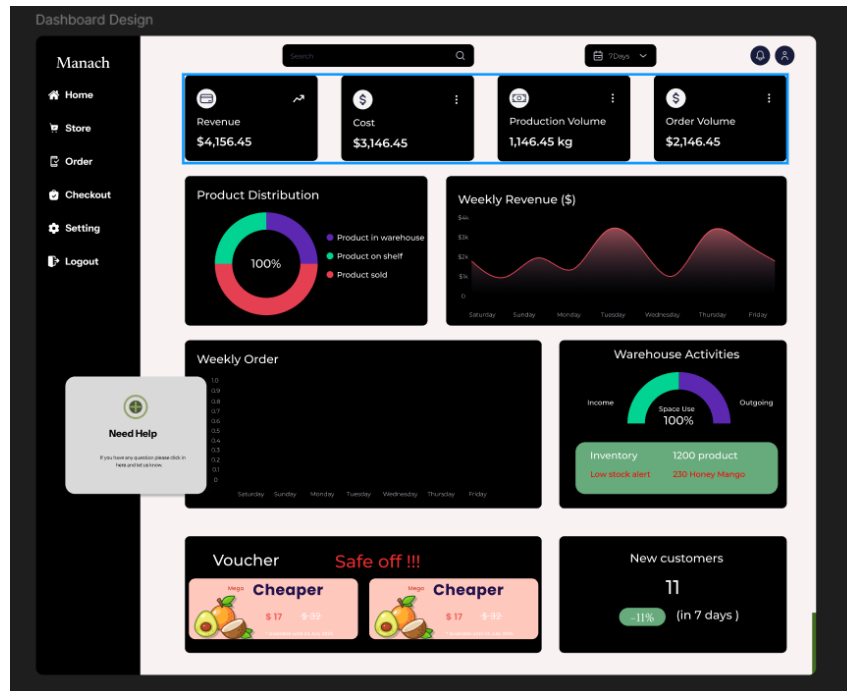


Figure 10 – Expected Dashboard page

Login

Email

Password

Log in

Don't have an account? [Register](#)

Register

Email

Full name

Password

Birth date

Phone number

City

Sign up

Already have an account? [Login](#)

Figure 11 – Expected Login/Register page

2. **Performance Requirement**

The Dining Guide web application is expected to provide a responsive and efficient user experience across all devices and interactions. To ensure smooth functionality and user satisfaction, the system must meet specific performance benchmarks. These response requirements define the acceptable time limits for various operations, including:

- **Page Load Time:** The home page and restaurant listing pages should load within 2–3 seconds under normal network conditions.
- **API Response Time:** All API calls (e.g., login, fetch reviews, submit reviews) should return responses within 500ms on average.
- **Concurrent Users:** The system should support up to 50 concurrent users during peak access without significant degradation.
- **Database Queries:** Queries for restaurant details and reviews should be optimized to execute in under 100ms.

3. **Maintainability Requirement**

The Dining Guide web application should be designed and implemented with maintainability in mind to support future updates, feature enhancements, and bug fixes with minimal effort. Key maintainability requirements include:

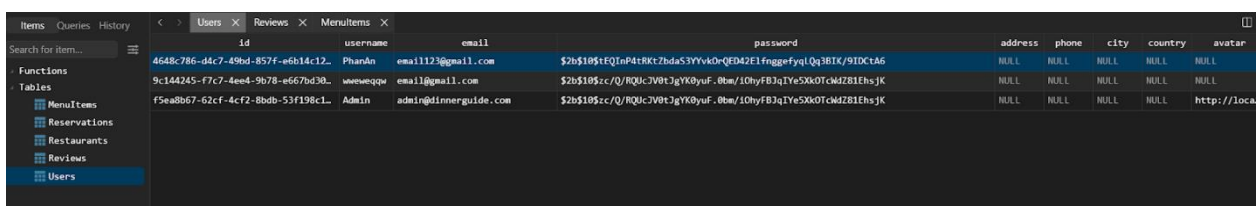
- **Modular Code Structure:** The codebase should follow a clear, modular architecture separating frontend, backend, and database logic for easier updates and debugging.
- **Readable Code:** Developers must follow consistent naming conventions, indentation, and commenting practices to ensure code clarity.

- Documentation: Essential components (e.g., API routes, database schemas, and main functions) must be documented for reference.
- Version Control: All development work should be managed via Git with clear commit messages and organized branching strategies.
- Scalability Consideration: The system should be built to allow easy addition of new features (e.g., tags, advanced filters, user badges) without major rewrites.

4. Security Requirements

To protect user data and ensure safe interactions within the Dining Guide web application, the system must adhere to essential security practices. These include:

- User Authentication & Authorization: Secure login and registration processes using hashed passwords (e.g., bcrypt) and session or token-based authentication (e.g., JWT).
- Input Validation & Sanitization: All user inputs (e.g., reviews, profile updates) must be validated and sanitized to prevent SQL injection and cross-site scripting (XSS) attacks.
- Secure Communication: The application should enforce HTTPS to ensure encrypted communication between client and server.
- Access Control: User roles and permissions should be enforced to restrict access (e.g., only authors can edit/delete their reviews).
- Session Management: Proper session timeout and logout features must be implemented to prevent unauthorized access from inactive sessions.



id	username	email	password	address	phone	city	country	avatar
4648c786-d4c7-49bd-857f-e6b14c12..	PhanAn	email123@gmail.com	\$2b\$10\$tEQInP4tRktZbda5YVvkdQED42ElfnggefyyLQq3BTK/9IDCtA6	NULL	NULL	NULL	NULL	NULL
9c144245-f7c7-4eed-9678-e667bd3b..	wwweweqw	email@gmail.com	\$2b\$10\$zc/Q/RQUC3V8t3gYK0yuf.0bm/10hyF8Jq1Ye5XkOTcm4281EhsJK	NULL	NULL	NULL	NULL	NULL
f5ea8b67-62cf-4cf2-8bdb-53f198c1..	Admin	admin@dinnerguide.com	\$2b\$10\$zc/Q/RQUC3V8t3gYK0yuf.0bm/10hyF8Jq1Ye5XkOTcm4281EhsJK	NULL	NULL	NULL	NULL	http://loca

Figure 12 – Successfully implemented database for security

II. Development process

The development of the dining guide website was guided by the **Model-View-Controller (MVC)** architecture to ensure a clear separation of concerns and maintainability. The environment consisted of the following key components:

Front-end

- **ReactJS** was utilized to build the user interface, enabling a responsive and interactive user experience. Its component-based architecture allowed for reusable UI elements and efficient state handling.

Back-end

- **Node.js** handled server-side operations and user request processing. Its non-blocking, asynchronous architecture made it ideal for scalable and efficient performance under concurrent usage.
- **Express.js**, a lightweight framework for Node.js, simplified the creation and management of RESTful APIs and server routes.

Database

- **MySQL** served as the primary data store, managing user information, restaurant reviews, and related content with reliable performance and structured querying capabilities.
- **Docker** was used to containerize the MySQL database, allowing for consistent, isolated environments across development and deployment.
- **TablePlus** provided a graphical interface for managing and inspecting the database, streamlining data visualization and query execution.

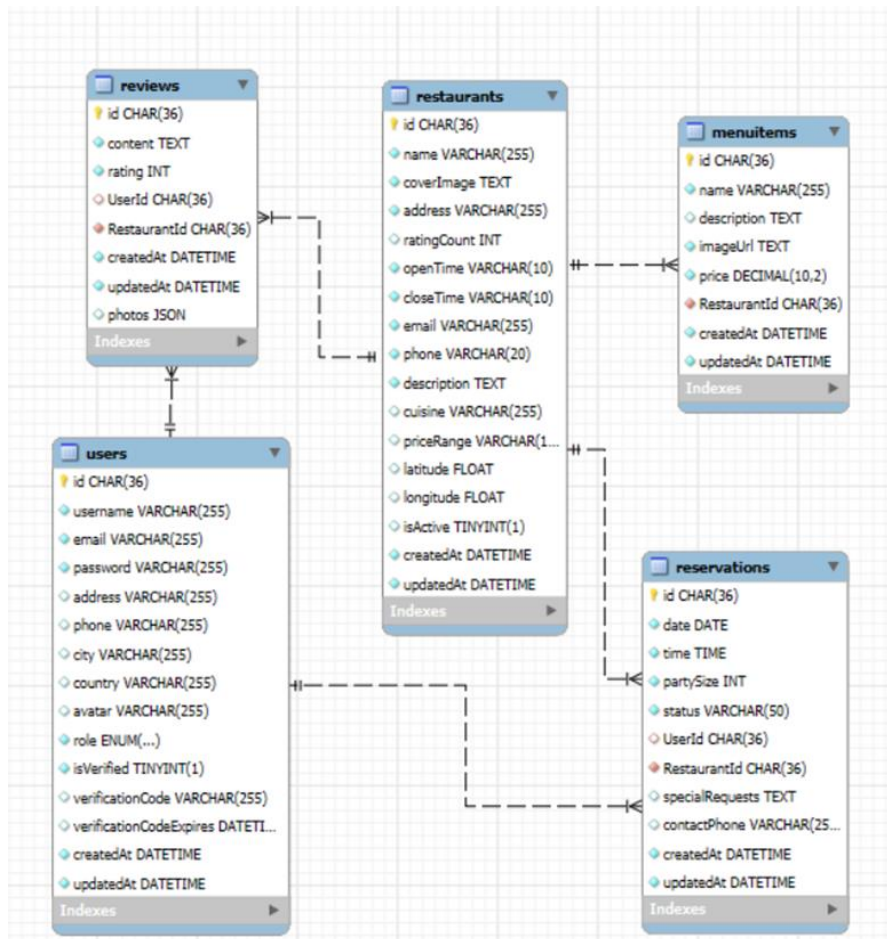


Figure 13 – Entity-Relationship Diagram in Dining Guide website

Development tools

- **Visual Studio Code** was used as the main integrated development environment (IDE) for coding and organizing project files.
- **Git & GitHub** facilitated version control and collaboration among team members, supporting continuous development and integration.

Architectural Pattern – MVC

The system was structured using the Model-View-Controller (MVC) design pattern:

- Model: Manages data stored in the MySQL database and accessed via backend logic.
- View: Handles presentation and user interaction via the ReactJS-based interface.

- c. **Controller**: Coordinates business logic, processes user input, and bridges data between the **Model** and **View**.

Work flow

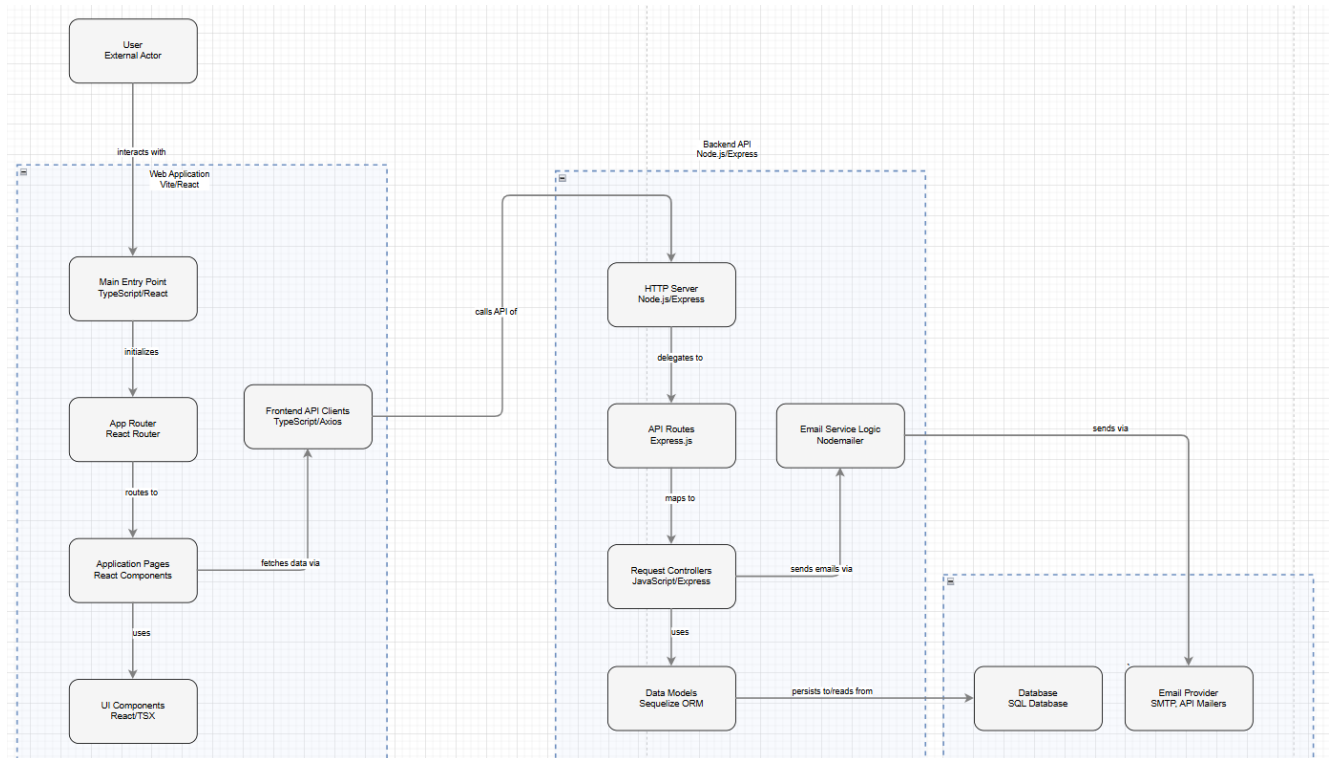


Figure 14 – Project work flow

The Dining Guide website follows a clear Model-View-Controller (MVC) architectural pattern to separate concerns and streamline development. The **user** interacts with the system through a web application built using **ReactJS** and **Vite**. The application initializes through a main entry point written in TypeScript and uses **React Router** to manage navigation between different pages. Each page consists of modular **React components** that render UI elements and handle interactions. To retrieve dynamic data, these components rely on **API clients** built with **Axios**, which communicate with the backend server.

On the server side, a **Node.js** application with **Express.js** handles incoming API requests. The Express HTTP server routes these requests to specific **API route handlers**, which are mapped to **controllers** responsible for processing the business logic. These controllers interact with the **Sequelize**

ORM to perform read and write operations on a **MySQL** database that stores key application data, such as user information and restaurant reviews. The database is containerized using **Docker**, ensuring consistent behavior across environments, and is managed via **TablePlus** for ease of inspection and querying.

In addition to database operations, the backend also includes an **email service** implemented using **Nodemailer**. When necessary, such as during user actions that require email notifications, the controllers trigger this service. The emails are sent via an external **SMTP or email API provider**, ensuring reliable delivery. This structure, as illustrated in the system diagram, ensures a smooth and maintainable development process, with the **model** managing data, the **view** presenting it to the user, and the **controller** coordinating the flow between them.

Use case diagram

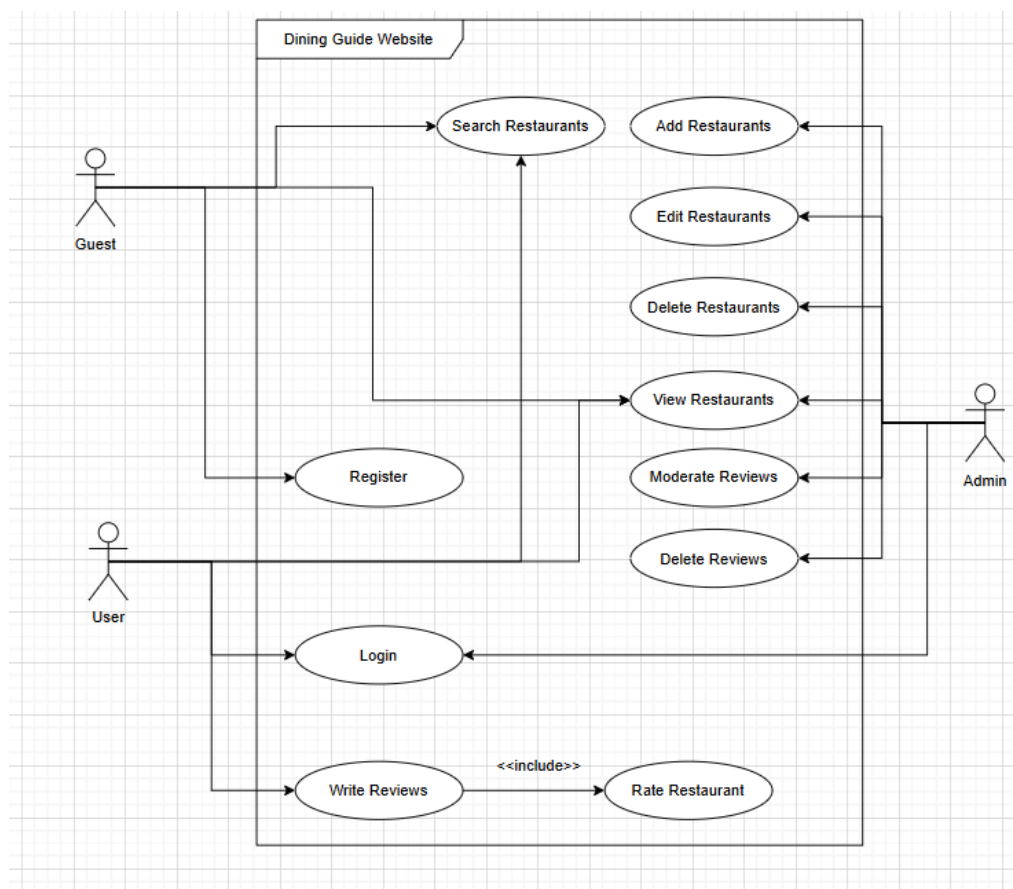


Figure 15 – Use case diagram

The use case diagram illustrates the core functionalities of the **Dining Guide Website**, identifying interactions between the system and three primary actors: **Guest**, **User**, and **Admin**.

A **Guest** can access the site to **search restaurants**, **view restaurant details**, and **register** for a user account. Once registered and logged in, a **User** gains additional capabilities such as **writing reviews**, which includes the option to **rate restaurants**.

An **Admin** has elevated privileges that extend beyond regular users. Admins can **add**, **edit**, and **delete restaurants**, as well as **moderate** and **delete user reviews**, ensuring content quality and compliance.

CHAPTER III

IMPLEMENTATION

I. User's flow

1. Homepage

The Homepage of Dining Guide website consists of five main sections:

a. Navigation bar:

- Left Side: Displays the logo of the website, Diningning.
- Right Side: Contains navigational buttons allowing users to move to different pages of the website and a button labeled either “Log In” or “Profile” depending on the user’s session status.

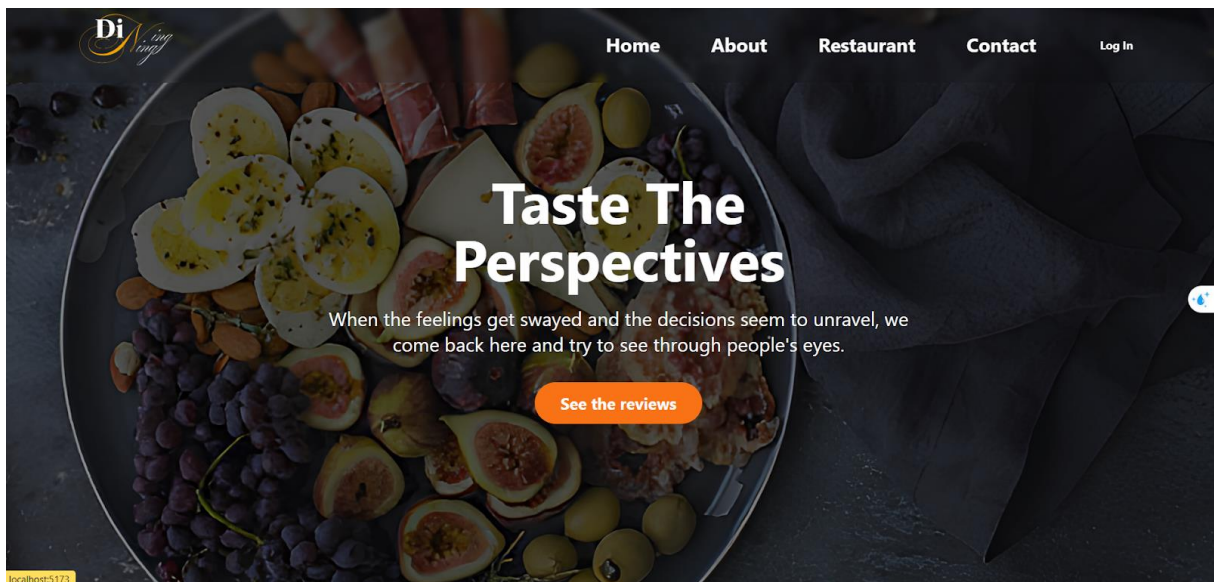


Figure 16 – Navigation bar section in Homepage

b. Website criteria:

This section states our website criteria using famous words from an influencing reviewer.

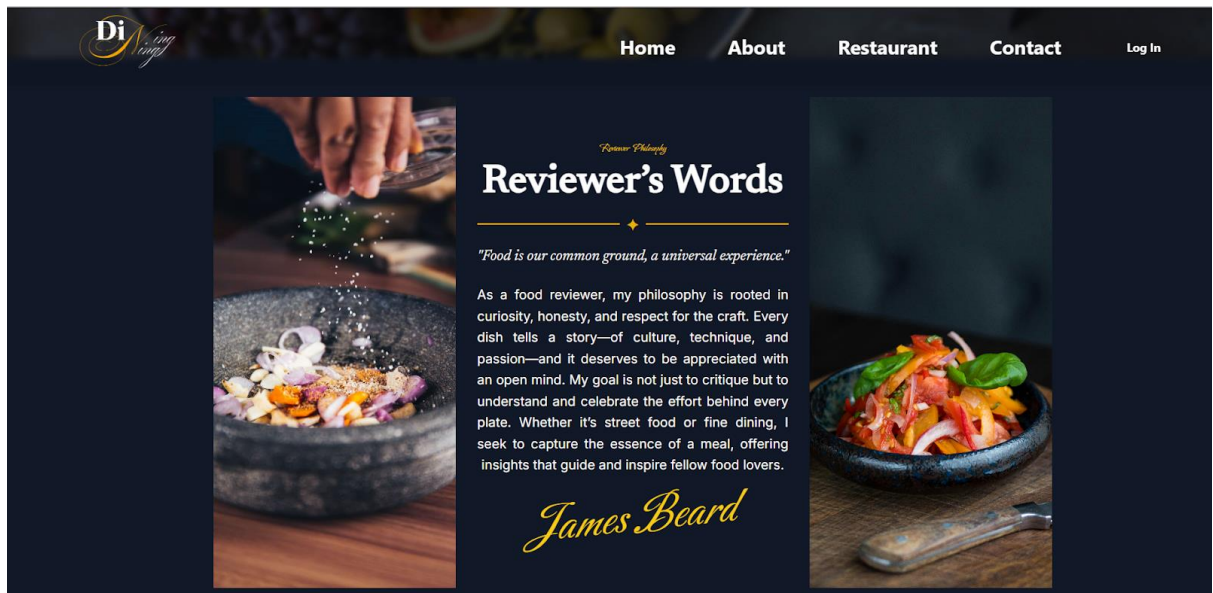


Figure 17 – Criteria section in Homepage

c. Featured pictures:

This section highlights prominent pictures from restaurants. Users can proactively drag the slide to see more pictures.

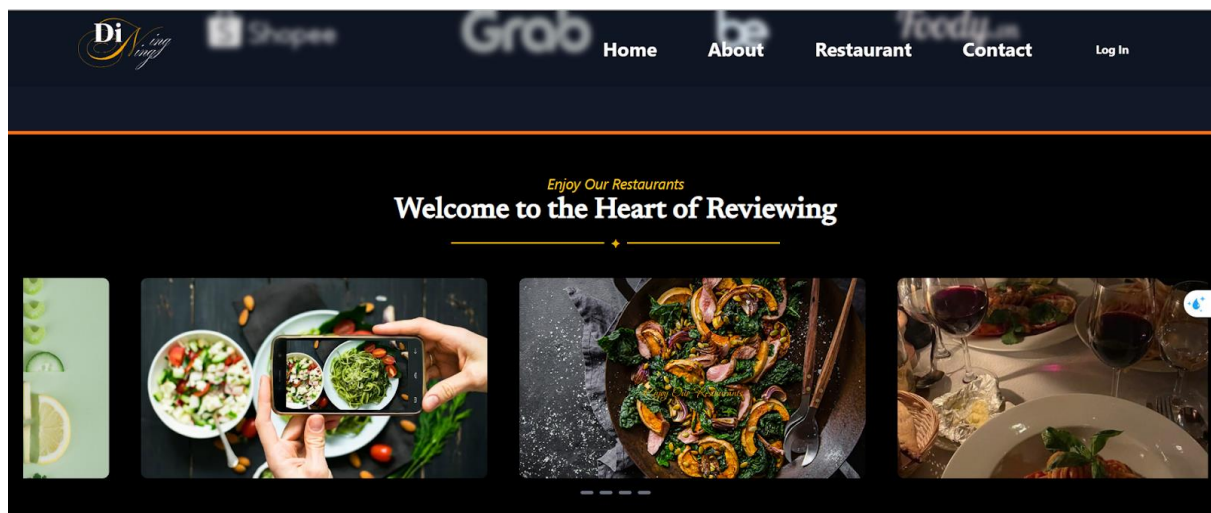


Figure 18 – Featured pictures in Homepage

d. Featured reviews:

This section includes prominent reviews from our previous users that improve the restaurants' fame and quality.

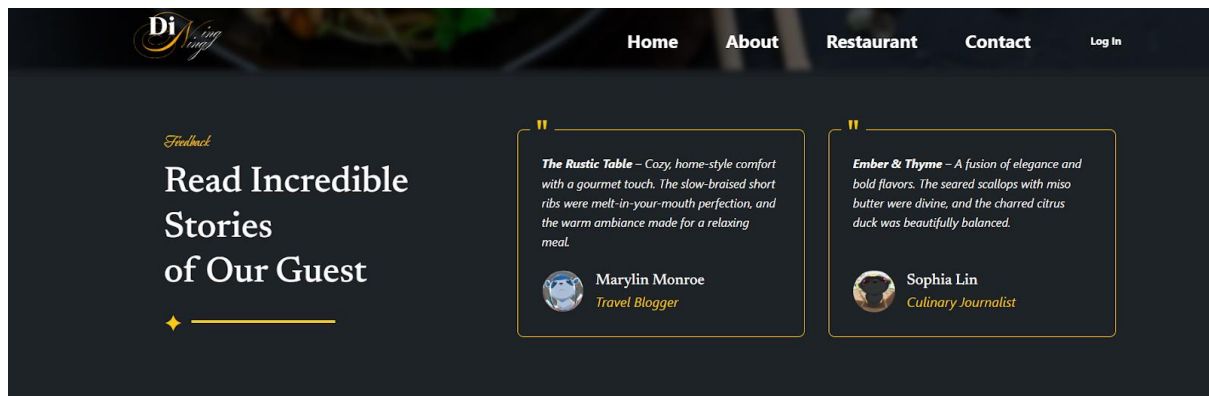


Figure 19 – Featured reviews in Homepage

e. Footer:

This section includes the logo of the website, second statement and buttons that can redirect users to the website's social network.

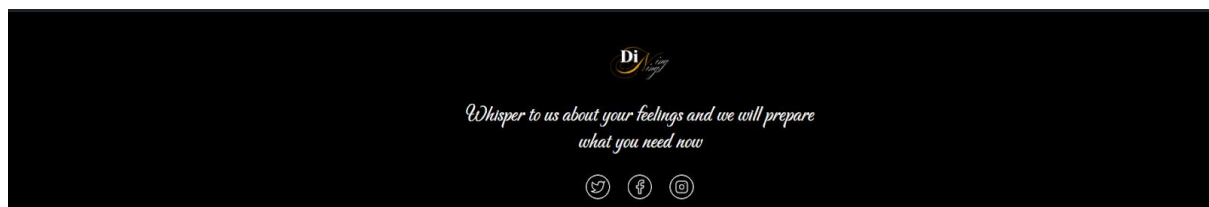
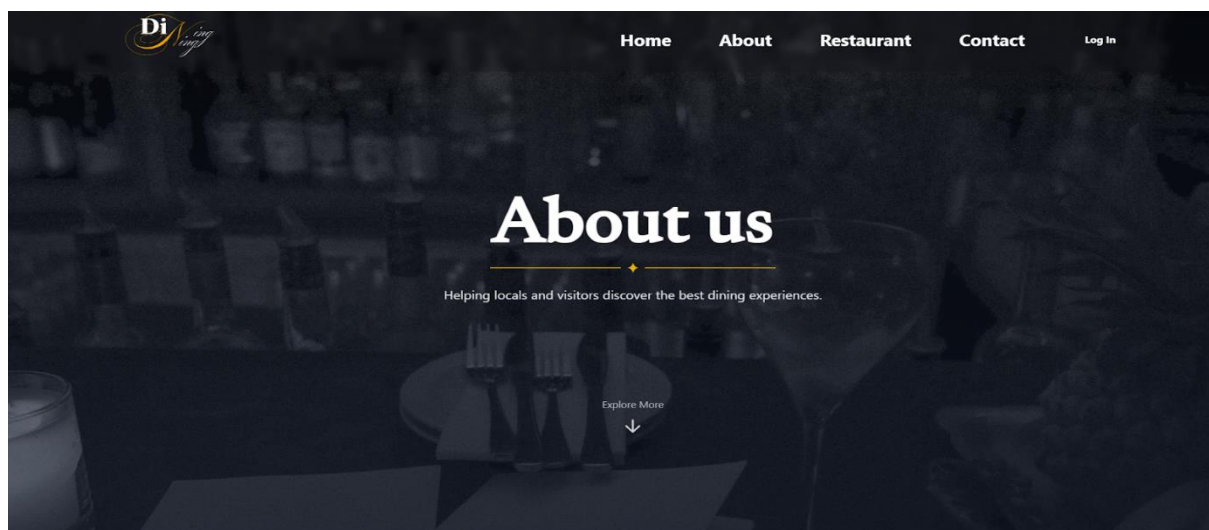


Figure 20 – Footer in Homepage

2. About us

The page is made to provide to the users the insights of this project, show the dedication about food from our perspective.



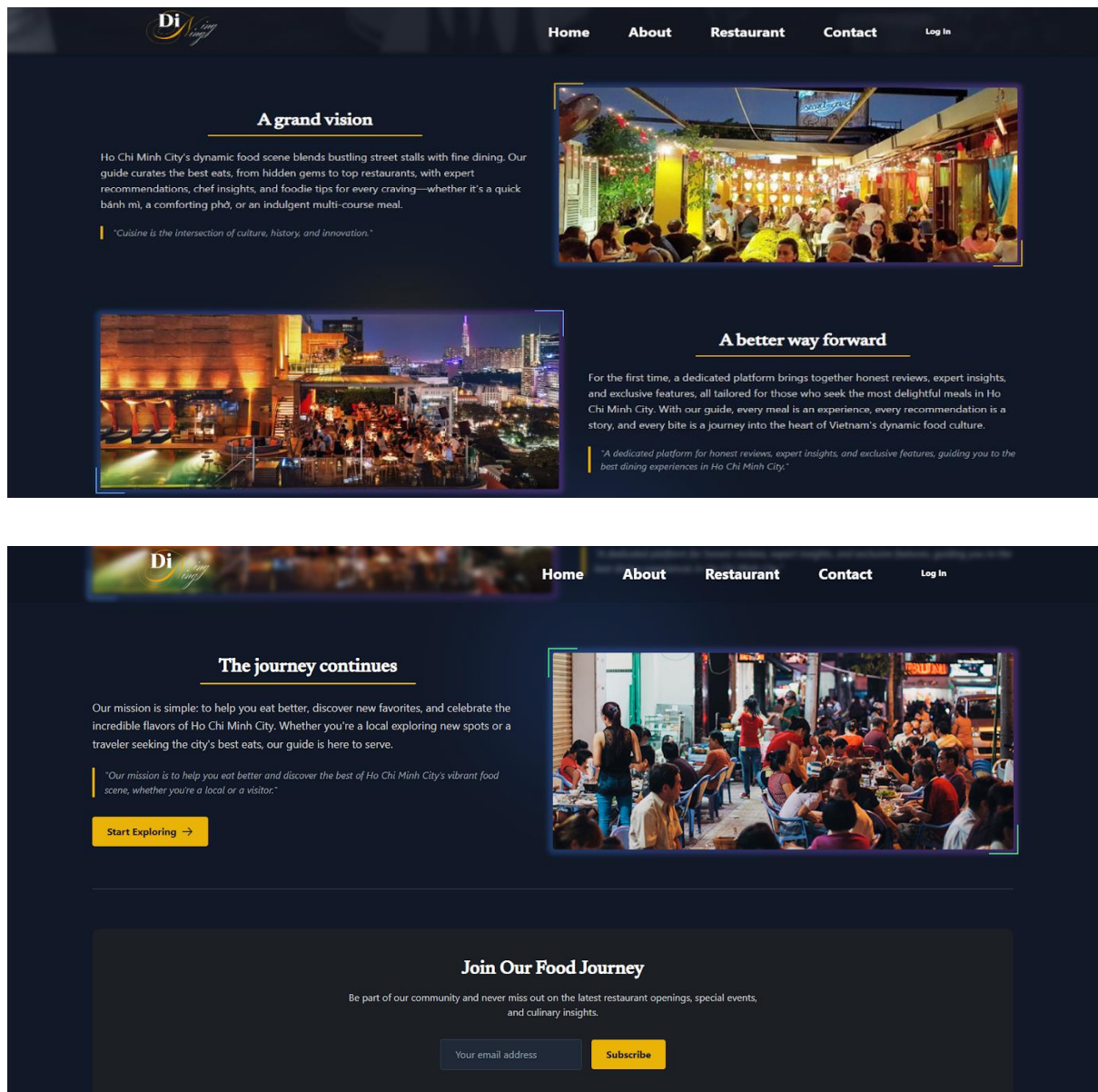


Figure 21 – About us page

3. Restaurant

In this Restaurant page of Dining Guide, users functionally can be the part of building the website using functions that implemented in this page.

Users can view the list of restaurants. Each restaurant card has its basic information displaying briefly. The cards can be sorted by using the sorting bar above.

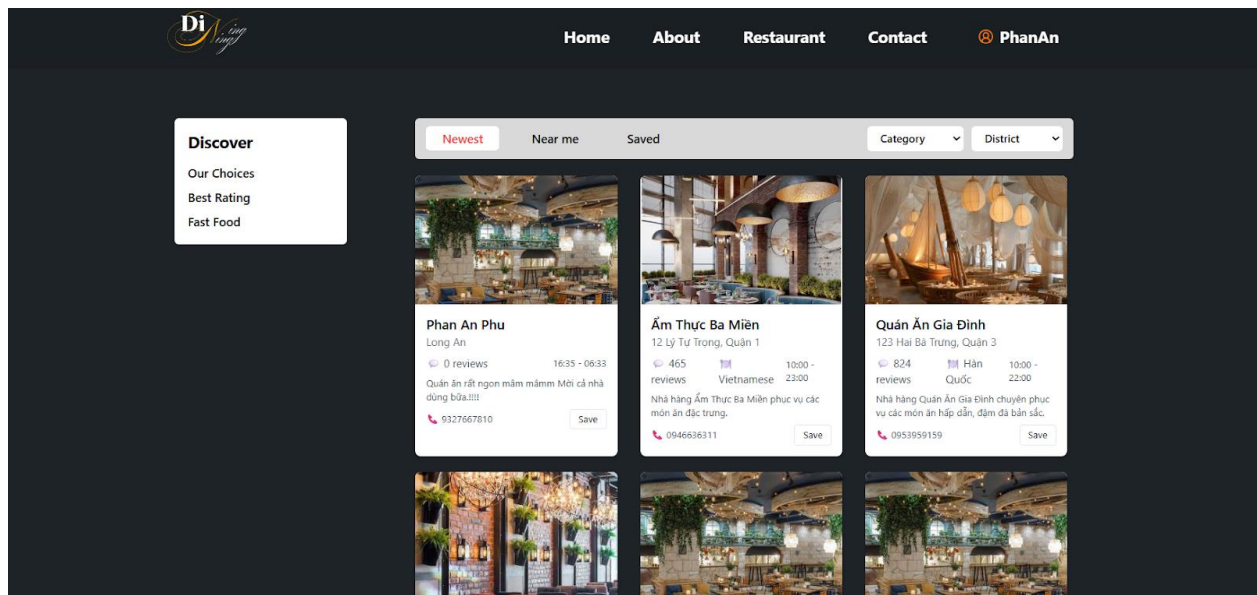
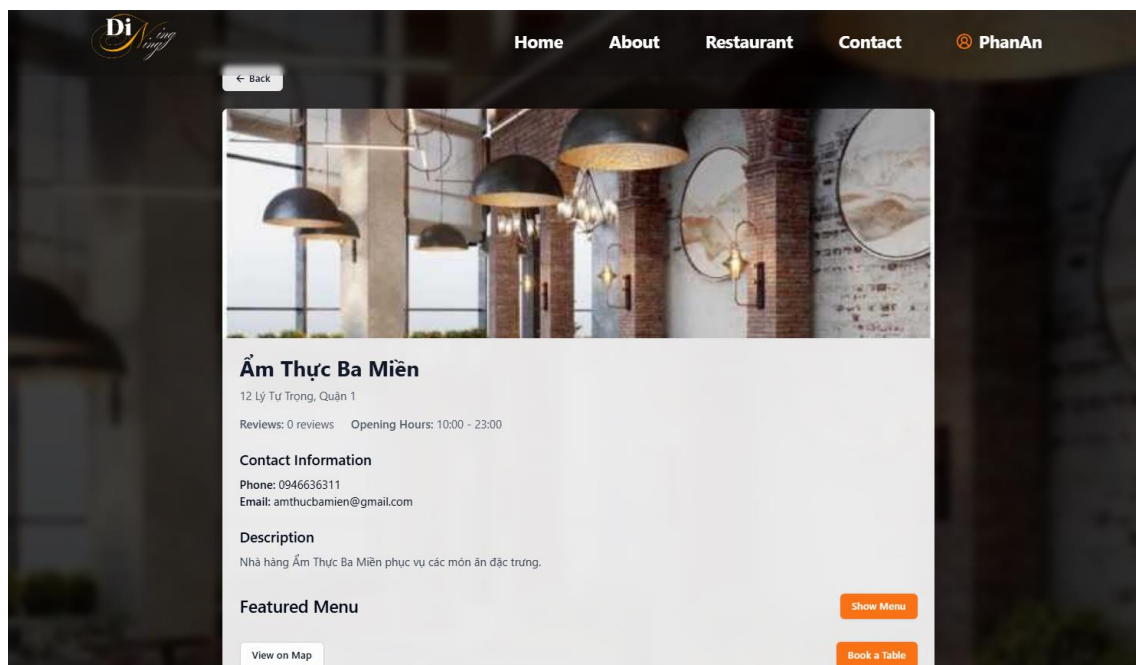


Figure 22 – Restaurant cards with sorting in Restaurant page

By clicking on restaurant card, users now can read the detailed information of that restaurant including Name, Address, Contact Information, e.t.c... The users can also see the menu, and search for this restaurant on Map. At the end of the card, there are reviews that can be written by authorized users.



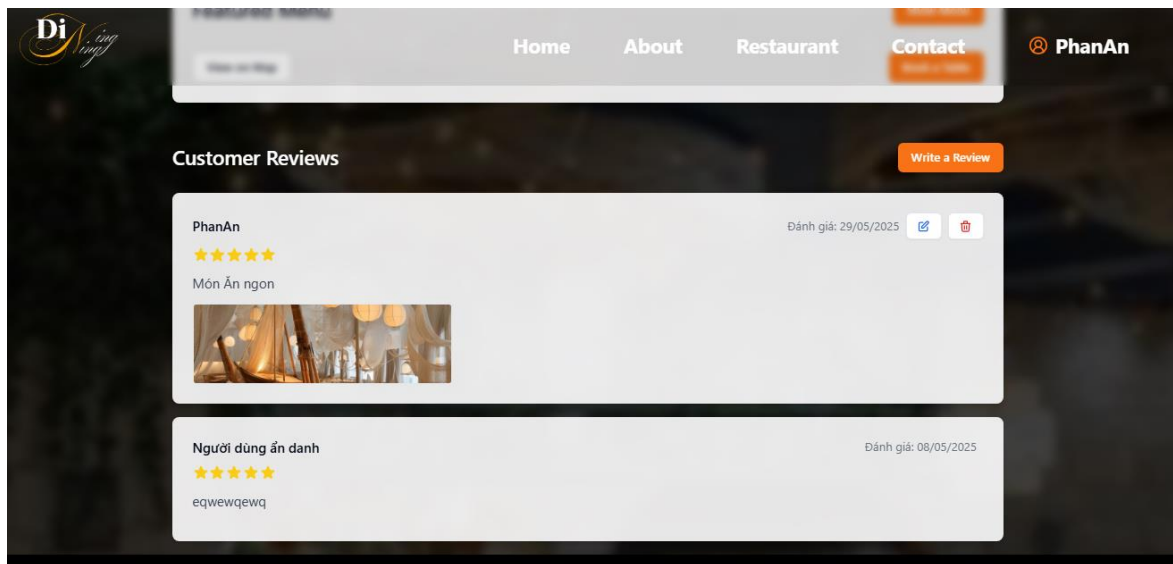


Figure 23 – Detailed restaurant card in Restaurant page

The registered users can write down their own opinions, take part in making the website completed.

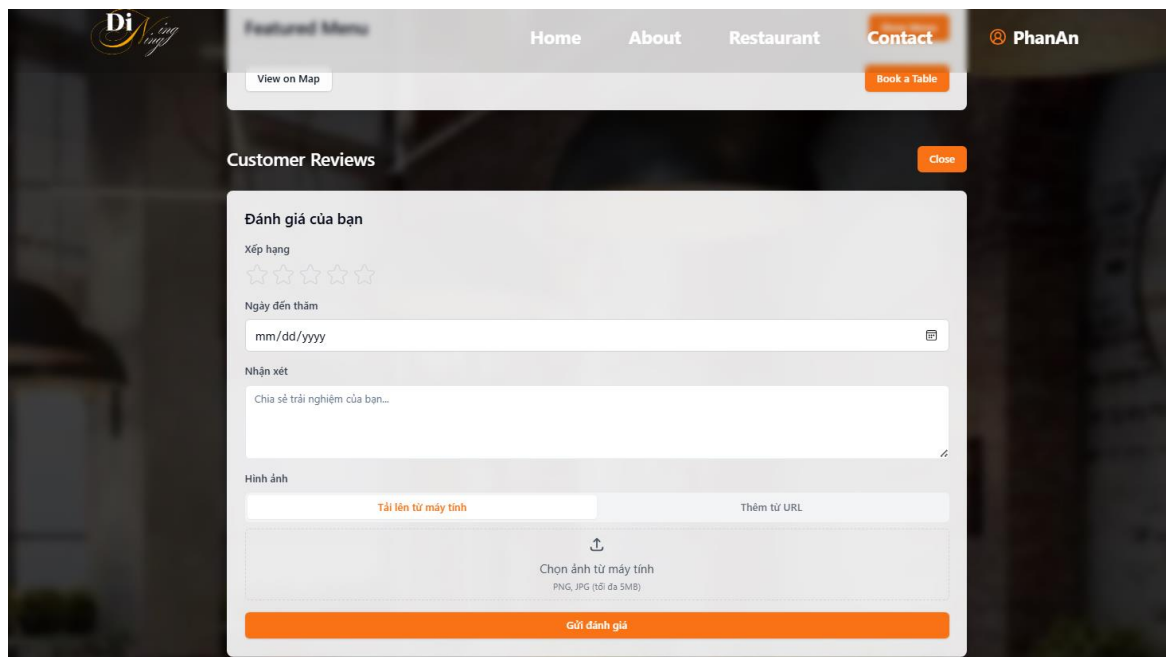


Figure 24 – Review writing feature in Restaurant page

4. Contact us

In this page, the users can give feedbacks privately to the developer via call, email or sending form. The feedbacks could be about the visual, how the website should work, etc...

Figure 25 – Contact us page

5. Login/Register

This section includes two main features:

a. Login

In this page, in order to access other features in the Restaurant page such as writing reviews, rating restaurant,... the users need to log in by inputting their credential stored in the database. If the users don't have account, they need to register by clicking the highlighted Register button.

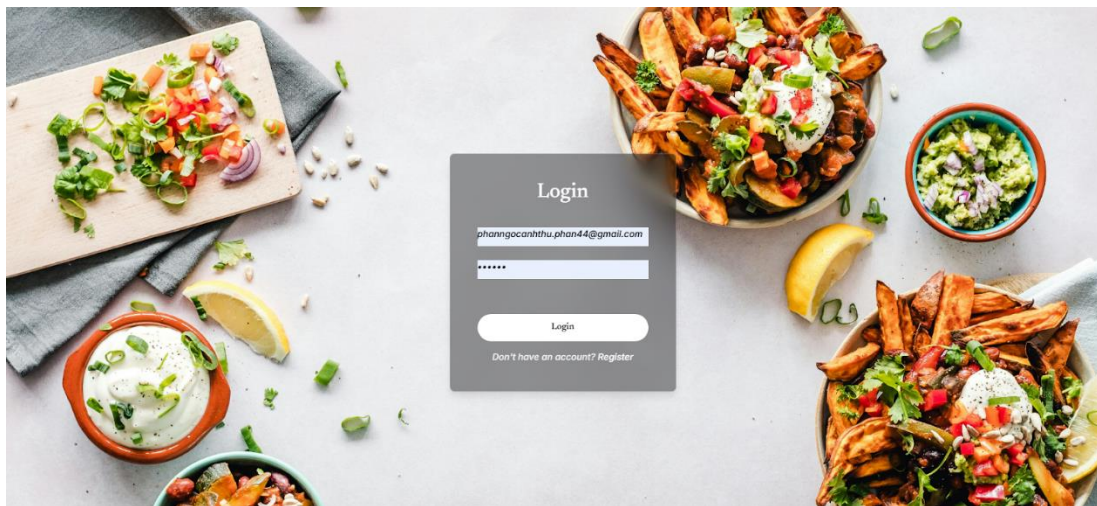


Figure 26 – Login page

b. Register

The users can sign up an account by inputting the required information. The account would be used for the login phase.

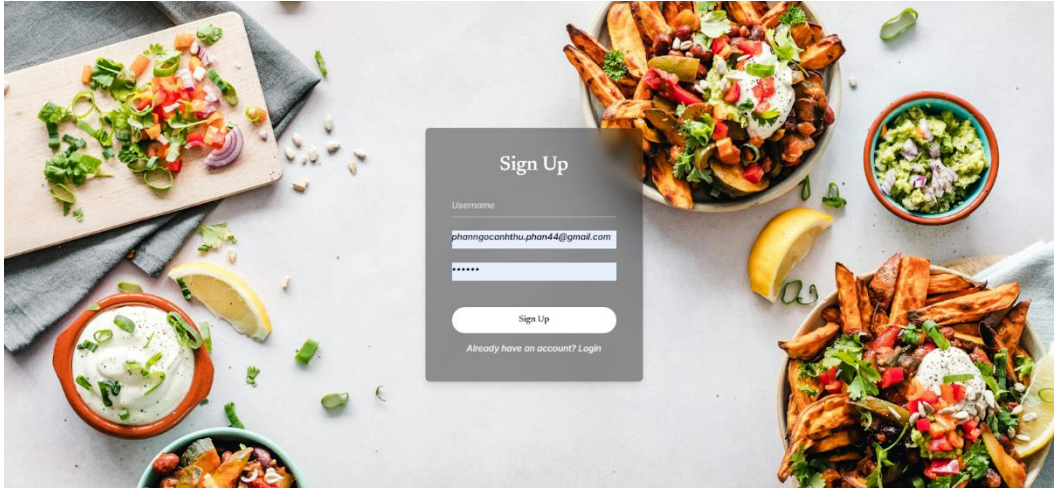


Figure 27 – Register page

II. Admin's flow

1. Homepage

The Admin Homepage of Dining Guide website consists of three main sections:

a. Header:

- Left Side: Displays the name of the website, Diningning.
- Middle Section: Contains a search bar to search for features, products, settings and more
- Right Side: Contains the avatar of the admin and its dropdown menu for settings.

b. Side bar:

- Top Side: Contains navigational buttons for redirecting to Homepage, Restaurants page, Dashboard page, Settings page, Review page.
- Bottom Side: Contains Help statement telling admin to seek support from the developer, and Logout button.

c. Main section:

- Contains navigational buttons for redirecting to Restaurants page, Dashboard page, Settings page.

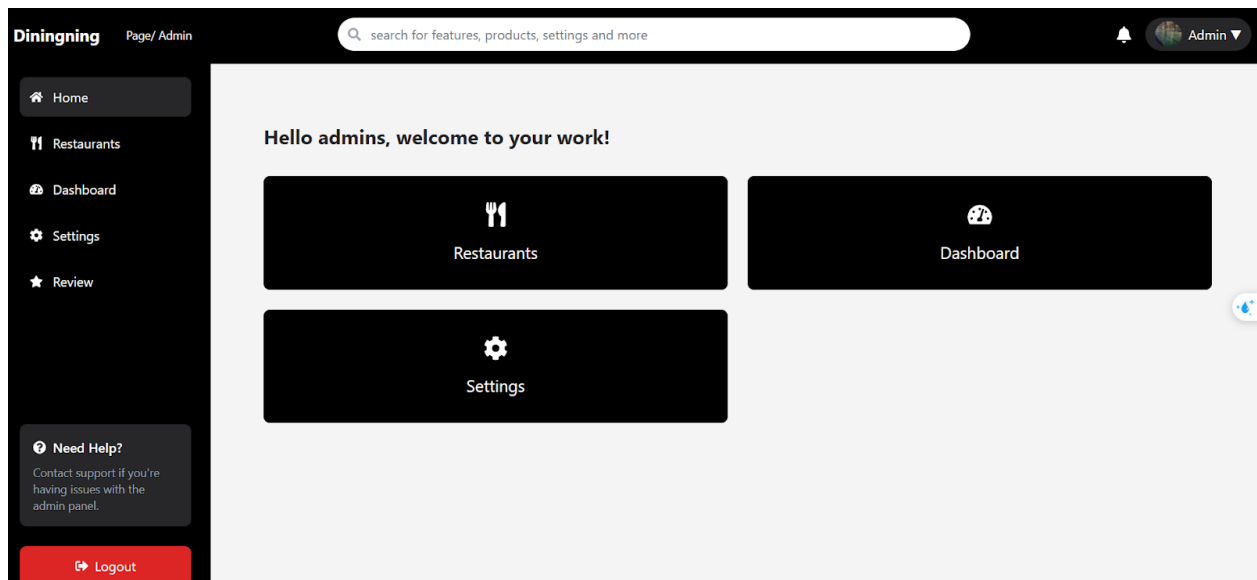


Figure 28 – Admin Homepage

2. Restaurants

In this Restaurant page of Dining Guide, admin can view the list of available restaurants and proactively add restaurants.

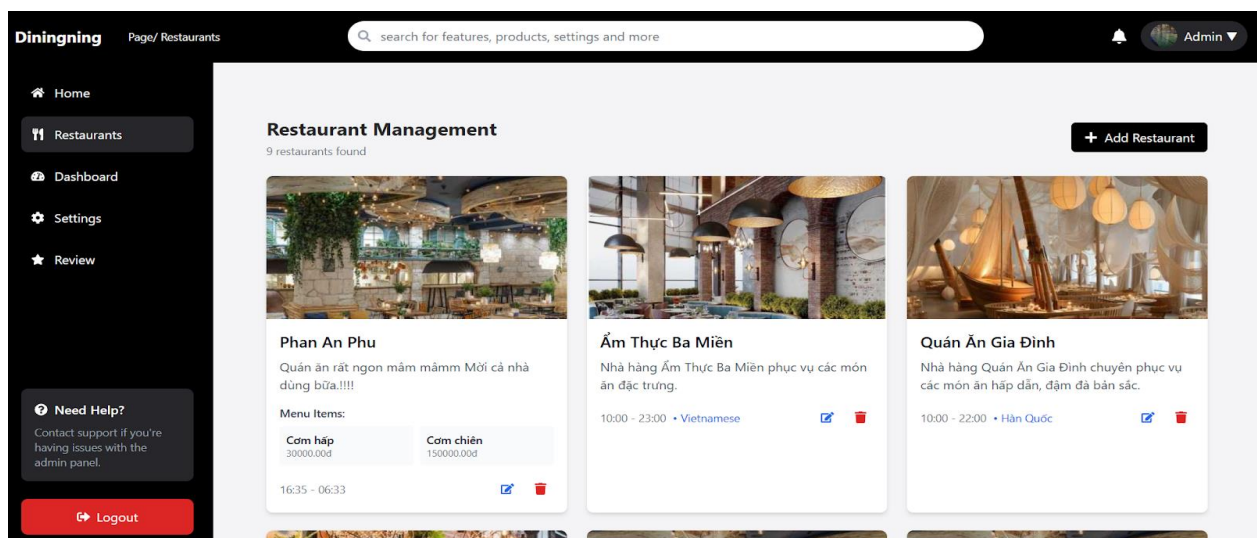


Figure 29 – Admin Restaurants Page

After clicking the “Add Restaurant”, admin can input required information for the card. The card after saved would appear on the user’s Restaurant page.

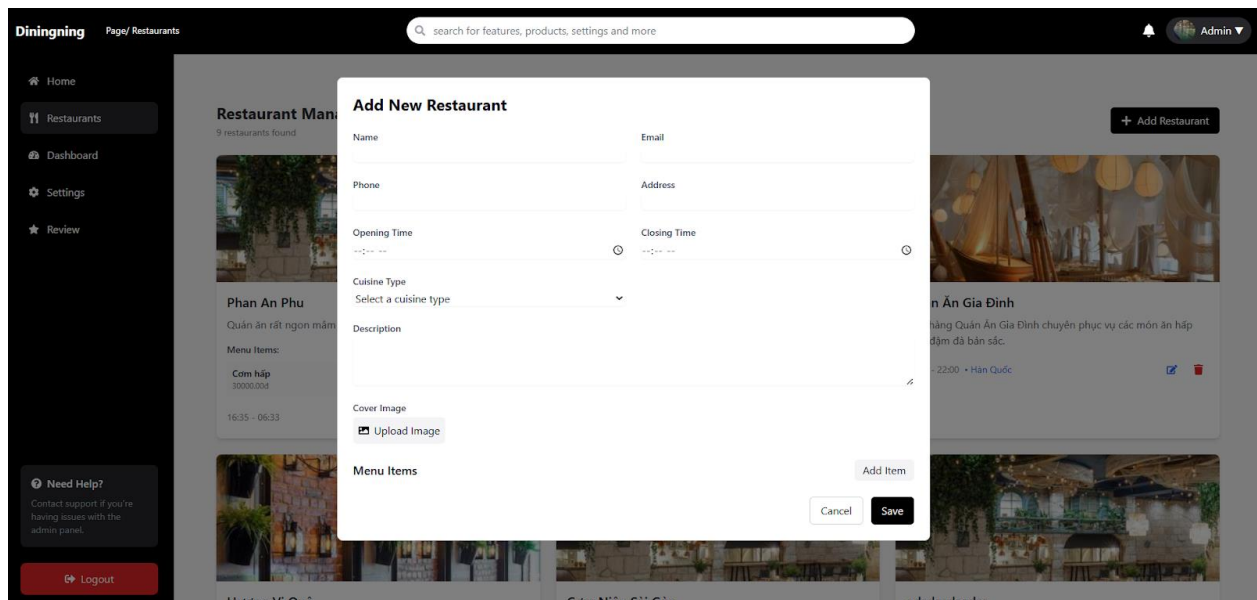


Figure 30 – Add Restaurant feature in Admin Restaurant page

The added restaurant card can be edited by clicking on it to reveal the modification box.

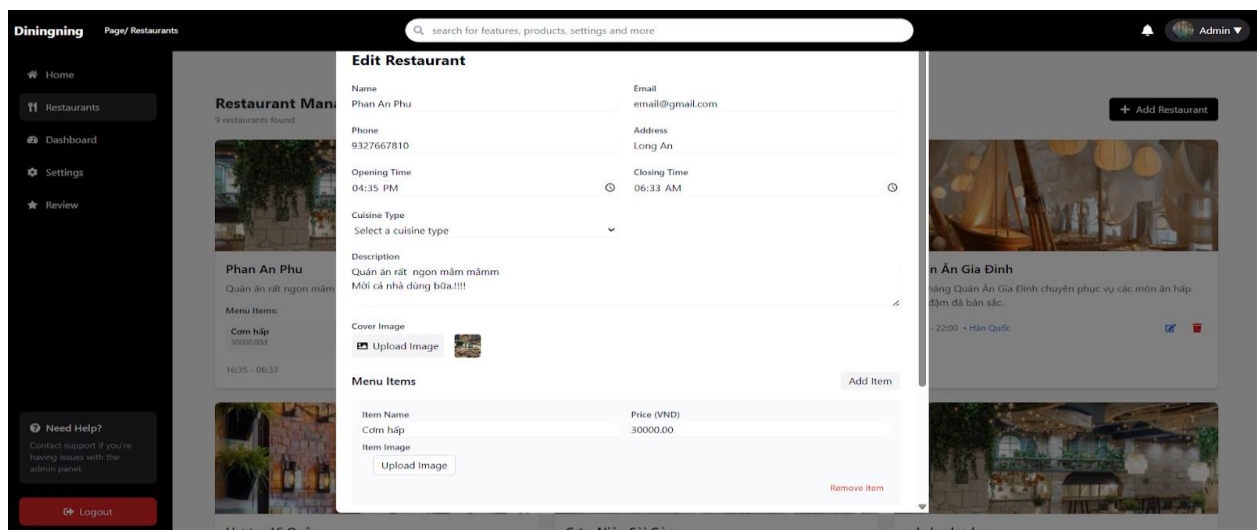


Figure 31 – Restaurant Modification feature in Admin Restaurant page

3. Dashboard

The Admin Dashboard is a centralized interface designed for managing the content and user activities within the Dining Guide web application. It allows administrators to monitor user reviews, manage restaurant listings, handle reported content, and oversee user accounts.

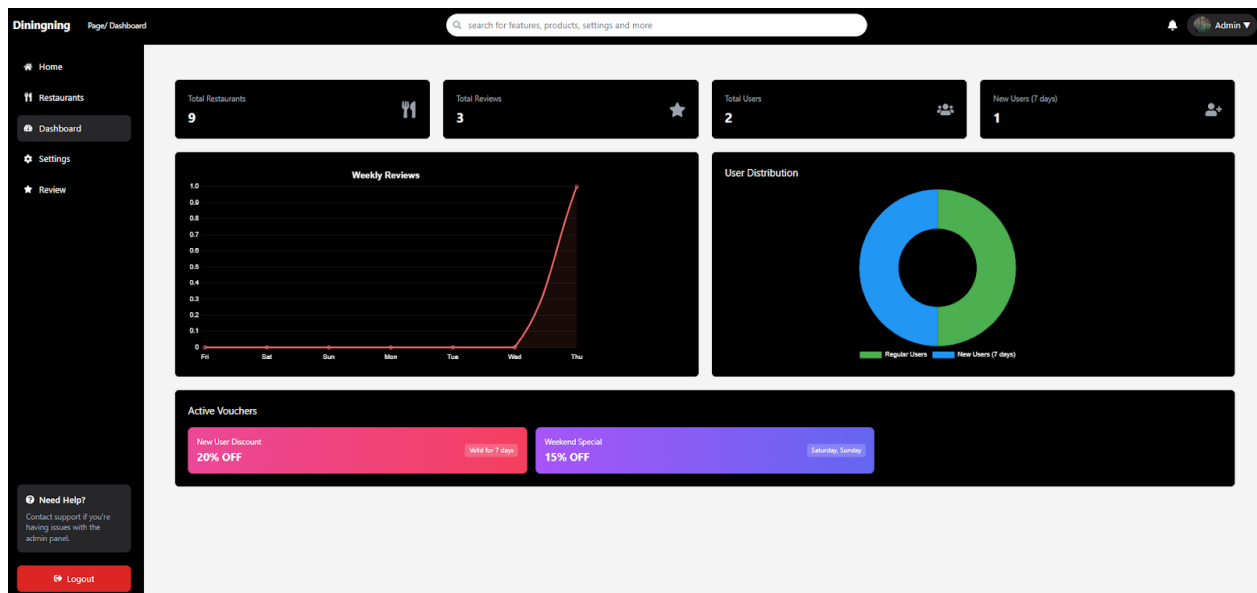


Figure 32 – Dashboard page

4. Settings

The Admin Settings section provides administrators with tools to customize and configure key aspects of their account. This includes managing admin profiles.

Figure 33 – Admin Settings page

5. Review

The Review Management feature allows administrators to oversee all user-submitted reviews on the Dining Guide platform. Admins can view, or delete reviews to ensure content quality and compliance with community guidelines. This system also includes tools for handling reported reviews,

detecting inappropriate content, and maintaining a respectful, trustworthy environment for all users. It plays a crucial role in upholding the platform's credibility and user experience.

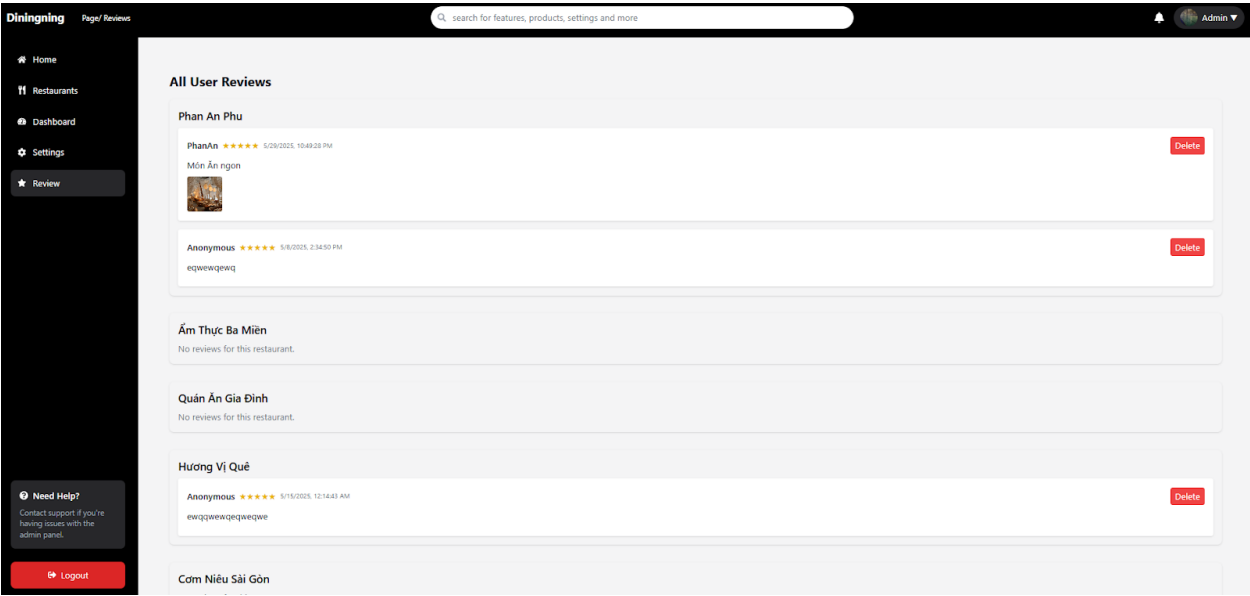


Figure 34 – Review page

CHAPTER IV

CONCLUSION

I. Discussion

This project introduced and developed a dynamic web-based Dining Guide platform, aimed at allowing users to discover, share, and manage restaurant reviews. Through a combination of modern web technologies—including ReactJS, Node.js, Express, and MySQL—the system delivers a responsive and interactive user experience. Core features such as user authentication, review posting, admin control, and content moderation were successfully implemented. The MVC architecture and containerization via Docker further enhanced the maintainability and scalability of the application.

The development process highlighted the importance of clear role distribution, iterative design, and continuous feedback. The collaboration between frontend and backend teams allowed for seamless integration of features and a consistent user interface. Implementing admin tools, such as review management and settings customization, ensured robust control over content quality. The use of Docker and version control systems (Git & GitHub) streamlined deployment and collaboration. Overall, the project not only met its functional goals but also laid a solid foundation for future enhancements, including recommendation features, AI support chat.

II. Future Improvements

One promising direction for future development is the integration of an **AI-powered support chat** feature. This virtual assistant would be available to guide users in real-time—helping them navigate the platform, find restaurants based on preferences, understand how to post reviews, or answer frequently asked questions. By using natural language processing (NLP), the AI chat can offer personalized suggestions, troubleshoot common issues, and enhance overall user satisfaction. This addition would reduce admin workload, improve accessibility, and create a more interactive and intelligent user experience on the Dining Guide platform.

III. References

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